Create 2 ubuntu instances

1.controller

2. server

Connect to server

Ubuntu machine default comes with python 3

But we need pytho 2

Connect to instance Info

Connect to your instance i-04b64e3a3b6529cf7 (server1) using any of these options

EC2 Instance Connect | Session Manager | SSH client | EC2 serial console

Instance ID

i-04b64e3a3b6529cf7 (server1)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is mahi18.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
   chmod 400 "mahi18.pem"
4. Connect to your instance using its Public DNS:
   ec2-3-110-103-61.ap-south-1.compute.amazonaws.com

Example:

ssh -i "mahi18.pem" ubuntu@ec2-3-110-103-61.ap-south-1.compute.amazonaws.com

Update apt repository

sudo apt-get update

Install python2

sudo apt-get install -y python2.7 python-pip

Establish password less connection between controller and managing nodes server

sudo passwd ubuntu (give password as ubuntu)

Sudo vim /etc/ssh/sshd_config

```
ubuntu@ip-172-31-5-200:~$ sudo vim /etc/ssh/sshd_config
ubuntu@ip-172-31-5-200:~$
```

```
# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
#PermitEmptyPasswords no
```

Change password authentication to yes

Save and quit

Restart the service
sudo service ssh restart

```
ubuntu@ip-172-31-5-200:~$ sudo service ssh restart
ubuntu@ip-172-31-5-200:~$
```

Exit

CONTROLLER

Now connected to controller

Update the repository
sudo apt-get update

Install python 2

Generate the keys
ssh-keygen

```
ubuntu@ip-172-31-4-180:~$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_rsa
Your public key has been saved in /home/ubuntu/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:2JP4spYOpoL2k7+1YTRtsiNsocxF7jwE7gkQFzsejqM ubuntu@ip-172-31-4-180
The key's randomart image is:
+---[RSA 3072]----+
|. o.             |
| o .             |
|. +. .           |
|.+.o+  +..       |
|o.o. =o+So       |
|..= O *.=.       |
|E. *.@.o.        |
|...oo Bo+        |
|. .oo+o.         |
+----[SHA256]-----+
ubuntu@ip-172-31-4-180:~$
```

Controller machine open

vim .ssh/id_rsa.pub

KEY EXAMPLE :


```
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABgQCuh9cGldjC/v5o6Tchw+VRlX2VWcUReTaaSqYUAiEgKvtW
O1ObQYfYHR+sgMOv2pqs/4M5m586od9CY2n3LPpSUKdj5ImDqiNOSaQEaRmNFBrCtPOtxo7n6urO
34pEY7Bhnu1oEJ2w7PboOzN5M+xBAi3wFVxHY9EdtccxgqODiBZOYOQps8EBhmYaZZ4SuPkUYEEv
hHtRfqG8AeexDm8WPRHGbkfwMFO9yN63XqKMyRwOMWWQmBXaycguEwoBo6sDopMQrwrlU8jSaJaA
HqQiHe8buPB3nTpX4DrBSozz7w3K9lxobUMyknDsFUqUTWIR3ndaS123+NruMNBmfoMxvt5myMOi
Q8taynhvsIh8sY+MzoDE996Un5WLhbLIGbdbg2IVS89VKU4pgr3Cssqa2a5DUMvQPy4YDlIo/Agh
cSOnVnTh6vf8yQfbQRnSBuPZGXIFaQqMgaOlvKVhjpvUS5irg84pNxfcu1wwshJK2MASQJgbaXuk
2PAMxUDegbM= ubuntu@ip-172-31-16-14
```


Exit

Connect to server1:

vim .ssh/authorized_keys --> paste the above key

Exit

Connect to controller machine

ssh -v ubuntu@server1 ip address

Connect to controller:

Install the ansible
sudo apt-get install software-properties-common

```
ubuntu@ip-172-31-4-180:~$ sudo apt-get install software-properties-common
Reading package lists... Done
```

sudo apt-add-repository ppa:ansible/ansible

```
0 upgraded, 0 newly installed, 0 to remove and 13 not upgraded.
ubuntu@ip-172-31-4-180:~$ sudo apt-add-repository ppa:ansible/ansible
Repository: 'deb https://ppa.launchpadcontent.net/ansible/ansible/ubuntu/ jam
```

sudo apt-get update

sudo apt-get install -y ansible

```
ubuntu@ip-172-31-4-180:~$ sudo apt-get install -y ansible
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Write the ip adresses of managing nodes in the inventory file

Cd /etc/ansible

$ls

Sudo vim hosts (copy paste server1 and 2 ip addresses)

//SETUP COMPLETED

Adhoc commands: it works based on modules

Command module:

Create files in server1 from controller

ansible all -i/etc/ansible/hosts -m command -a 'touch f1 f2 f3'

```
ubuntu@ip-172-31-4-180:/$ ansible all -i/etc/ansible/hosts -m command -a 'touch f1 f2 f3'
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/py
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendic
172.31.5.200 | CHANGED | rc=0 >>

[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/py
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendic
172.31.7.139 | CHANGED | rc=0 >>
```

To check files are created or not

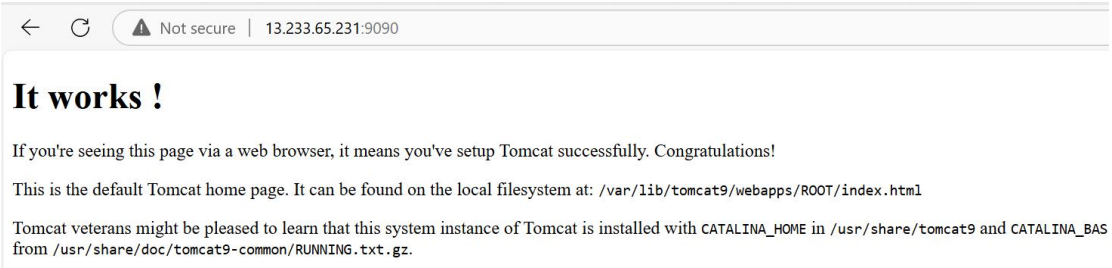
Ssh 172.31.5.200

$ls

```
ubuntu@ip-172-31-4-180:/$ ssh 172.31.5.200
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1021-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Jan 22 15:27:28 UTC 2025

  System load:  0.08               Processes:             102
  Usage of /:   28.4% of 7.57GB    Users logged in:       0
  Memory usage: 24%                IPv4 address for eth0: 172.31.5.200
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
13 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

6 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


Last login: Wed Jan 22 15:26:15 2025 from 172.31.4.180
ubuntu@ip-172-31-5-200:~$ ls
f1  f2  f3  file1  file2  file3
ubuntu@ip-172-31-5-200:~$
```

Install tomcat9

ansible all -m apt -a 'name=tomcat9 state=present update_cache=yes' -b

```
ubuntu@ip-172-31-4-180:/$ ansible all -m apt -a 'name=tomcat8 state=present update_cache=yes' -b
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python in
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
172.31.7.139 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "msg": "No package matching 'tomcat8' is available"
}
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python in
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
172.31.5.200 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "msg": "No package matching 'tomcat8' is available"
}
ubuntu@ip-172-31-4-180:/$ ansible all -m apt -a 'name=tomcat9 state=present update_cache=yes' -b
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python in
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
172.31.5.200 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "cache_update_time": 1737560819,
    "cache_updated": true,
    "changed": true,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "Reading package lists...\nBuilding dependency tree...\nReading state information...\nThe following packages were automatically installed and ar
quired:\n  git-man liberror-perl\nUse 'sudo apt autoremove' to remove them.\nThe following additional packages will be installed:\n  alsa-topology-conf alsa-u
tificates-java default-jre-headless\n  java-common libapr1 libasound2 libasound2-data libavahi-client3\n  libavahi-common-data libavahi-common3 libcups2 libec
-java\n  libgraphite2-3 libharfbuzz0b liblcms2-2 libpcsclite1 libtcnative-1\n  libtomcat9-java openjdk-11-jre-headless tomcat9-common\nSuggested packages:\n
basound2-plugins alsa-utils cups-common liblcms2-utils pcscd\n  libnss-mdns fonts-dejavu-extra fonts-ipafont-gothic fonts-ipafont-mincho\n  fonts-wqy-microhei
enhei fonts-indic tomcat9-admin tomcat9-docs\n  tomcat9-examples tomcat9-user\nThe following NEW packages will be installed:\n  alsa-topology-conf alsa-ucm-co
ates-java default-jre-headless\n  java-common libapr1 libasound2 libasound2-data libavahi-client3\n  libavahi-common-data libavahi-common3 libcups2 libeclipse
\n  libgraphite2-3 libharfbuzz0b liblcms2-2 libpcsclite1 libtcnative-1\n  libtomcat9-java openjdk-11-jre-headless tomcat9 tomcat9-common\n0 upgraded, 22 newly
```

To check take the server1 publicip:8080